

Que 1

PRIME OR NOT

Aim:

To write a shell script to check whether a given number is prime or not.

PROGRAM:

```
echo "Enter a number: "  
read num  
i=2  
f=0  
while [ $i -le `expr $num / 2` ]  
do  
if [ `expr $num % $i` -eq 0 ]  
then  
f=1  
fi  
i=`expr $i + 1`  
done  
if [ $f -eq 1 ]  
then  
echo "The number is composite"  
else  
echo "The number is Prime"  
fi
```

OUTPUT:

```
Enter a number:  
5  
The number is Prime
```

Ex.NO:2**DISPLAY VALID LOGIN NAME****Aim:**

To write a shell script to accept a login name. If not a valid login name display message entered login name is invalid.

Program:

```
#!/bin/sh
name="admin"
echo "Enter Your Role"
read name1
if [ $name = $name1 ]
then
    echo "You are admin"
else
    echo "You are not admin"
fi
```

OUTPUT:

```
Enter Your
Role admin
You are admin
```

Que 3 DATE

DISPLAY

Aim:

To Write a shell script to display date in the mm/dd/yy format..

Program:

```
#!/bin/bash
CURRENTDATE=`date +"%Y-%m-%d %T"`
CURRENTDATEONLY=`date +"%b %d, %Y"`
CURRENTEPOCHTIME=`date +"%Y-%m-%d %T"`
echo Current Date is: ${CURRENTDATEONLY}
echo Current Date and Time is: `date +"%Y-%m-%d %T"`
echo Current Date and Time is: ${CURRENTDATE}
echo Current Unix epoch time is: ${CURRENTEPOCHTIME}
```

OUTPUT:

```
Current Date is: Mar 06, 2019
Current Date and Time is: 2019-03-06 17:08:23
Current Date and Time is: 2019-03-06 17:08:23
Current Unix epoch time is: 2019-03-06 17:08:23
```

Ex.NO 4

MULTIPLICATION TABLE

Aim:

To write a shell script to display the multiplication table any number.

Program:

```
clear
echo enter a number
read n
for i in {1..10}
do
    echo "$i*$n"=$((i * $n))
done
```

OUTPUT:

```
enter a number
2
1*2= 2
2*2= 4
3*2= 6
4*2= 8
5*2= 10
6*2= 12
7*2= 14
8*2= 16
9*2= 18
10*2= 20
```

Ex.NO:5 FILE NAME AS INPUT AND DISPLAY WHETHER IT EXISTS OR NOT

Aim:

To write a shell script to accept a file name as input and display whether it exists or not.

Program:

```
clear
echo "Enter the filename"
read name
echo
echo "File details"
if [ -f $name ]
then
echo "File exists"
echo -n "file size is:"
wc -c $name
echo -n "File type:"
file $name
if [ -r $name -a -w $name -a -x $name ]
then
echo "File is readable writeable and executable"
elif [ -r $name -a -w $name ]
then
echo "File is readable and writable"
elif [ -r $name ]
then
echo "File is read only"
elif [ -w $name ]
```

```
then
echo "File is write only "
elif [ -x $name ]
then
echo "File executable only"
else
echo "File not readable ,writable and executable"
fi
else
echo "File does not exist"
fi
```

OUTPUT:

Enter the filename

file1

File details

File exists

file size is:27 file1

File type:file1: ASCII text

File is readable and writable

Ex.NO:6
NUMBERS**SUM OF THE FIRST N****Aim:**

To write a shell script to find the sum of the first n numbers.

Program:

```
echo "Enter a number: "  
read num  
  
i=1  
sum=0  
  
while [ $i -le $num ]  
do  
sum=`expr $sum + $i`  
i=`expr $i + 1`  
done  
echo "The sum of first $num numbers is: $sum"
```

OUTPUT:

```
Enter a number:  
5  
The sum of first 5 numbers is: 15
```

Ex.NO:7
FILE.

COMPARING AND SORTING THE

Aim:

To write a shell script to merge the contents of three files, sort the contents and then display them page by page

Program:

```
echo "***Comparing and sorting the file***"  
echo "first file name with extension "  
read f1  
echo "enter second file name with extension "  
read f2  
comm <(sort f1.txt) <(sort f2.txt)
```

OUTPUT:

first file name with extension

f1.txt

enter second file name with extension

f2.txt

ai are

doing

happen hi

what

Ex.NO:8
CALCULATOR

Aim:

To write a shell script to stimulate a calculator that does the addition, subtraction, multiplication and division between two numbers. The user should be prompted for two operands and operator.

Program:

```
#!/bin/bash

# Take user Input
echo "Enter Two numbers : "
read a
read b
# Input type of operation
echo "Enter Choice : "
echo "1. Addition"
echo "2. Subtraction"
echo "3. Multiplication"
echo "4. Division"
read ch
# Switch Case to perform
# calculator operations
case $ch in
1)res=`expr $a + $b`
;;
2)res=`expr $a - $b `
;;
3)res=`expr $a \* $b`
;;
4)res=`expr $a / $b`
;;
esac
echo "Result : $res"
```

Ex.NO:9**FIND THE POWER OF GIVEN NUMBER****Aim:**

To write a shell script that takes two numbers through keyboard and finds the value of one number raised to the power of another.

Program:

```
echo "Enter a number: "  
read a  
  
echo "Enter Power: "  
read p  
  
i=1  
ans=1  
while [ $i -le $p ]  
do  
ans=`expr $ans \* $a`  
i=`expr $i + 1`  
done  
echo "Answer of $a^$p is $ans"
```

OUTPUT

```
Enter a number: 10  
Enter Power: 3  
Answer of 10^3 is 1000
```

Ex.NO:10 FIND THE LARGEST OF THREE NUMBERS

Aim:

To find the largest of three numbers and also find the total and the average

Program:

```
echo "Enter the 1st Number: "  
read n1  
echo "Enter the 2nd Number: "  
read n2  
echo "Enter the 3rd Number: "  
read n3  
total=`expr $n1 + $n2 + $n3`  
avg=`expr $total / 3`  
if [ $n1 -gt $n2 -a $n1 -gt $n3 ]  
then  
echo "$n1 is the Largest of three"  
elif [ $n2 -gt $n1 -a $n2 -gt $n3 ]  
then  
echo "$n2 is the Largest of three"  
elif [ $n3 -gt $n1 -a $n3 -gt $n2 ]  
then  
echo "$n3 is the Largest of three"  
else  
echo "All the three numbers are Equal"  
fi  
echo "Total: $total"  
echo "Average: $avg"
```

OUTPUT

```
Enter the 1st Number:  
5  
Enter the 2nd Number:  
6  
Enter the 3rd Number:  
7  
7 is the Largest of three  
Total: 18  
Average: 6
```

Ex.NO:11

FIND A FACTORIAL

Aim:

To Write a script to find a factorial of a given number.

Program:

```
fact=1
ie=1

echo -e "Enter a number:\c"
read a

while [ $ie -le $a ]
do
    fact=`expr $fact \* $ie`
    ie=`expr $ie + 1`
done
echo -e "Multilpication of $a number is $fact."
```

OUTPUT

```
Enter a number:10
Multilpication of 10 number is 3628800.
```